

Homework #2] Gerardo Gutierrez

PROBLEM #1

Prove Gauss-Markov Theorem, i.e. show that the least squares estimate in linear regression is the BLUE (Best Linear Unbiased Estimate), which means $\text{Var}(a^T \hat{\beta}) \leq \text{Var}(c^T y)$ where $c^T y$ is any unbiased estimator for $a^T b$.

$\Rightarrow$ suppose we have

$$y = X\beta + \epsilon$$

where $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$ is observed training data labels
 $X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}$ is $\mathbb{R}^{n \times p+1}$ n rows = data points & p cols = features

and $\beta = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_p \end{bmatrix}$ is the regression coefficients used to minimize error in order to predict.

ϵ is the "act of God" noise in the data
where $\epsilon \sim N(0, \sigma^2)$ $\epsilon = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}$

$\Rightarrow y, X$ are known training data. ϵ is error that is fixed but not known.

we want to estimate β using $\hat{\beta}$. We want to show the least square linear estimates is BLUE.

$\Rightarrow$ Though we already showed least square estimator of β is $\hat{\beta} = (X^T X)^{-1} X^T y$, we will reshow it.

given $\hat{y} = X\beta$ where $\hat{y}$ is prediction & X is test data, we derive the error as

$$y - \hat{y} = y - X\beta \quad \begin{array}{l} y = \text{ground truth} \\ \hat{y} = \text{prediction.} \end{array}$$

$$e_i = y_i - \hat{y}_i \text{ error}$$

we do the sum of square errors which is

$$J(\beta) = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - X_{ij} \beta_j)$$

where $J(\beta)$ is the cost function we wish to minimize.

$$\begin{aligned} \frac{\partial J}{\partial \beta_k} = 0 &= \sum_{i=1}^n \frac{\partial}{\partial \beta_k} \left(y_i - \sum_{j=1}^n X_{ij} \beta_j \right)^2 \\ &= \sum_{i=1}^n 2 \left(y_i - \sum_{j=1}^n X_{ij} \beta_j \right) \frac{\partial}{\partial \beta_k} \left(- \sum_{j=1}^n X_{ij} \beta_j \right) \end{aligned}$$

$\frac{\partial \beta_j}{\partial \beta_k} = \delta_{jk}$

$$= \sum_{i=1}^n -2 \left(y_i - \sum_{j=1}^n x_{ij} \beta_j \right) \sum_{j=1}^n x_{ij} \delta_{jk}$$

$$= \sum_{i=1}^n -2 \left(y_i - \sum_{j=1}^n x_{ij} \beta_j \right) x_{ik}$$

$$= \sum_{i=1}^n \left[-2 y_i x_{ik} + 2 \sum_{j=1}^n x_{ij} \beta_j x_{ik} \right] = 0$$

$$\sum_{i=1}^n y_i x_{ik} = \sum_{i=1}^n \sum_{j=1}^n x_{ij} \beta_j x_{ik}$$

using einstein convention

$$y_i x_{ik} = x_{ij} \beta_j x_{ik}$$

$$x_{ki}^T y_i = x_{ki}^T x_{ij} \beta_j \quad \text{write in matrix form}$$

$$X^T X \hat{\beta} = X^T y$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

→ least square linear estimator of β when $y = X\beta + \epsilon$

now, we have to show $\hat{\beta}$ is BLUE, i.e., Best Linear Unbiased Estimator.

$\hat{\beta}$ is linear since of form $\hat{\beta} = A y$ $\rightarrow (X^T X)^{-1} X^T y$

$\hat{\beta}$ is unbiased. unbiased means $E(\hat{\beta}) - E(\beta) = 0$

$$E(\hat{\beta}) = E((X^T X)^{-1} X^T y)$$

$$= E((X^T X)^{-1} X^T (X\beta + \epsilon))$$

$$= E((X^T X)^{-1} X^T X \beta + X^T \epsilon)$$

$$= E((X^T X)^{-1} (X^T X) \beta) + E(X^T \epsilon)$$

$$= E(\beta) + E(X^T \epsilon)$$

Since X^T is known, not a r.v so can pull out of expectation.

$$= E(\beta) + X^T E(\epsilon)$$

$\epsilon \sim N(0, \sigma^2)$ is zero-mean so $E(\epsilon) = 0$

$$= E(\beta) + X^T \cdot 0$$

$$E(\hat{\beta}) = E(\beta) \Rightarrow E(\hat{\beta}) - E(\beta) = 0$$

hence $\hat{\beta}$ is unbiased.

$\hat{\beta}$ is Best Estimator. The "Best Estimator" has minimum variance. let's first calc $\text{Var}(\hat{\beta})$

$$\text{Var}(\hat{\beta}) = E(\hat{\beta}^2) - (E(\hat{\beta}))^2$$

* note, I calc'd $\hat{\beta}^2 \sim ((X^T X)^{-1} X) ((X^T X)^{-1} X)$ but the multiplication didn't work out. need to be careful w/ matrix multiplication. let's start w/ $E(\hat{\beta}^2)$ *

$$E(\hat{\beta}^2) = E(\hat{\beta} \hat{\beta}^T) \\ = E\left[(X^T X)^{-1} X^T y \right] \left[(X^T X)^{-1} X^T y \right]^T$$

* using $(ABC)^T = C^T B^T A^T$ *

$$= E\left[(X^T X)^{-1} X^T y \right] \left[y^T X (X^T X)^{-1} \right]^T$$

* using $(A^{-1})^T = (A^T)^{-1}$ *

$$= E\left[(X^T X)^{-1} X^T y \right] \left[y^T X (X^T X)^{-1} \right]^T \\ = E\left[(X^T X)^{-1} X^T y \right] \left[y^T X (X^T X)^{-1} \right]^T$$

$$= E\left[(X^T X)^{-1} X^T y y^T X (X^T X)^{-1} \right]$$

* now $y = X\beta + \epsilon$. $y^T = \beta^T X^T + \epsilon^T$

$$= E\left[(X^T X)^{-1} X^T (X\beta + \epsilon) (\beta^T X^T + \epsilon^T) X (X^T X)^{-1} \right]$$

$$= E\left[(X^T X)^{-1} X^T \left(X\beta\beta^T X^T + X\beta\epsilon^T + \epsilon\beta^T X^T + \epsilon\epsilon^T \right) X (X^T X)^{-1} \right]$$

$$= E\left[\left(\underbrace{(X^T X)^{-1} X^T X}_{\beta\beta^T X^T} \underbrace{\beta^T X^T}_{\beta^T X^T} + \underbrace{(X^T X)^{-1} X^T X}_{\beta\epsilon^T} \underbrace{\beta\epsilon^T}_{\beta\epsilon^T} + \underbrace{(X^T X)^{-1} X^T \epsilon}_{\epsilon\beta^T X^T} \underbrace{\beta^T X^T}_{\beta^T X^T} + \underbrace{(X^T X)^{-1} X^T \epsilon}_{\epsilon\epsilon^T} \right) X (X^T X)^{-1} \right]$$

$$= E\left[\left(\beta\beta^T X^T + \beta\epsilon^T + (X^T X)^{-1} X^T \epsilon\beta^T X^T + (X^T X)^{-1} X^T \epsilon\epsilon^T \right) X (X^T X)^{-1} \right]$$

$$= E\left[\beta\beta^T X^T X (X^T X)^{-1} + \beta\epsilon^T X (X^T X)^{-1} + (X^T X)^{-1} X^T \epsilon\beta^T X^T X (X^T X)^{-1} \right]$$

$$+ (X^T X)^{-1} X^T \epsilon \epsilon^T X (X^T X)^{-1}]$$

$$= E(\beta \beta^T + \beta \epsilon^T X (X^T X)^{-1} + (X^T X)^{-1} X^T \epsilon \beta^T + (X^T X)^{-1} X^T \epsilon \epsilon^T X (X^T X)^{-1})$$

*the only r.v is ϵ *

$$= \beta \beta^T + \beta X (X^T X)^{-1} \cancel{E(\epsilon^T)} + (X^T X)^{-1} X^T \beta^T \cancel{E(\epsilon)} + (X^T X)^{-1} X^T \epsilon \epsilon^T X (X^T X)^{-1} E(\epsilon \epsilon^T)$$

ϵ is zero mean

$$= \beta \beta^T + (X^T X)^{-1} E(\epsilon \epsilon^T) \quad \text{in matrix form}$$

$$E(\hat{\beta}^2) = \beta^2 + (X^T X)^{-1} E(\epsilon^2)$$

now $\text{Var}(\epsilon^2) = E(\epsilon^2) - (E(\epsilon))^2 = E(\epsilon^2)$ which is σ^2

$$= \beta^2 + (X^T X)^{-1} \sigma^2 \quad \text{back to } \text{Var}(\hat{\beta}^2)$$

$$\begin{aligned} \text{Var}(\hat{\beta}^2) &= E(\hat{\beta}^2) - (E(\hat{\beta}))^2 \\ &= \beta^2 + (X^T X)^{-1} \sigma^2 - (E(\hat{\beta}))^2 \end{aligned}$$

* we have already shown since $\hat{\beta}$ is unbiased
 $E(\hat{\beta}) = E(\beta) = \beta$

$$\text{Var}(\hat{\beta}) = \beta^2 + (X^T X)^{-1} \sigma^2 - \beta^2 = (X^T X)^{-1} \sigma^2$$

$$\text{so } \text{var}(\hat{\beta}) = (X^T X)^{-1} \sigma^2$$

for a generic unbiased linear estimator,

instead of $\hat{\beta} = (X^T X)^{-1} X^T y$ we let

$\tilde{\beta} = G y$ where $\tilde{\beta}$ is unbiased. so

$$\text{var}(\tilde{\beta}) = E(\tilde{\beta}^2) - (E(\tilde{\beta}))^2$$

$$E(\tilde{\beta}^2) = E(\tilde{\beta} \tilde{\beta}^T)$$

$$= E((Gy)(Gy)^T)$$

$$= E(G(X\beta + \epsilon)(X\beta + \epsilon)^T G^T)$$

$$= E(G(X\beta + \epsilon)(\beta^T X^T + \epsilon^T) G^T)$$

$$= E(G(X\beta + \epsilon)(\beta^T X^T + \epsilon^T) G^T)$$

$$= E(G [X\beta\beta^T X^T + X\beta\epsilon^T + \epsilon\beta^T X^T + \epsilon\epsilon^T] G^T)$$

$$= E(GX\beta\beta^T X^T G^T) + E(GX\beta\epsilon^T G^T) + E(G\epsilon\beta^T X^T G^T) + E(G\epsilon\epsilon^T G^T)$$

again, only r.v is ϵ

$$= GX\beta\beta^T X^T G^T + \cancel{GX\beta G^T E(\epsilon^T)} + \cancel{G\beta^T X^T G^T E(\epsilon)} + GG^T E(\epsilon\epsilon^T)$$

$$= GX\beta\beta^T X^T G^T + GG^T E(\epsilon\epsilon^T) \text{ rewrite as matrix}$$

$$E(\tilde{\beta}^2) = (GX\beta)^2 + G^2 E(\epsilon^2)$$

$$\begin{aligned}
 \text{Var}(\tilde{\beta}) &= E(\tilde{\beta}^2) - (E(\tilde{\beta}))^2 \\
 &= (G\beta)^2 + b^2 E(\varepsilon^2) - (E(\tilde{\beta}))^2 \\
 &\quad \text{since } \tilde{\beta} \text{ is unbiased } E(\tilde{\beta}) = E(\beta) = \beta \\
 &= (G\beta)^2 + b^2 E(\varepsilon^2) - \beta^2 \quad E(\varepsilon^2) = \sigma^2
 \end{aligned}$$

$$\text{Var}(\tilde{\beta}) = (G\beta)^2 + b^2 \sigma^2 - \beta^2$$

$$\begin{aligned}
 E(\tilde{\beta}) &= E(Gy) = E(G(X\beta + \varepsilon)) \\
 &= G\beta
 \end{aligned}$$

in order to satisfy unbiased estimator, we require
 $GX = I$ where I is the identity matrix. or $G = X^T$

so

$$\begin{aligned}
 \text{Var}(\hat{\beta}) &= (GX\beta)^2 - \beta^2 + b^2 \sigma^2 \\
 &= \beta^2 - \beta^2 + b^2 \sigma^2
 \end{aligned}$$

$$\text{Var}(\hat{\beta}) = b^2 \sigma^2$$

So for BLUE estimator $\hat{\beta}$: $\text{Var}(\hat{\beta}) = (X^T X)^{-1} \sigma^2$
 for generic unbiased linear estimator $\tilde{\beta}$: $\text{Var}(\tilde{\beta}) = b^2 \sigma^2$

Hw wants us to show $\text{Var}(\hat{\beta}) \leq \text{Var}(\tilde{\beta})$
this equates to comparing matrices σ^2 & $(X^T X)^{-1}$.
we wish to show $(X^T X)^{-1} \leq \sigma^2 = \sigma \sigma^T$
 $\Rightarrow$ to be unbiased, we required $\sigma X = \hat{1}$

$$\sigma \sigma^T - (X^T X)^{-1} \geq 0$$

we know $\hat{\beta} = (X^T X)^{-1} X^T y$ let $A = (X^T X)^{-1} X^T$
 $\tilde{\beta} = \sigma y$

$$\hat{\beta} = Ay \quad \text{and} \quad \tilde{\beta} = \sigma y$$

$\hat{\beta}$ & $\tilde{\beta}$ are unbiased so $AX = (X^T X)^{-1} X^T X = \hat{1}$
 $\sigma X = \hat{1}$

since both are unbiased, their difference is unbiased

$(\sigma - A)X$ should equal

$$(\sigma - A)X = \left((X^T X)^{-1} X^T - \sigma \right) X = \hat{1} - \hat{1} = 0.$$

so $(\sigma - A)X = 0$ let their difference be

$$D = \sigma - A. \quad \text{so} \quad DX = 0.$$

convert to G : $G = D + A$

now we have something for $GG^T - (X^T X)^{-1} \geq 0$

$$\begin{aligned} GG^T &= (D+A)(D+A)^T = (D+A)(D^T + A^T) \\ &= DD^T + DA^T + AD^T + AA^T \end{aligned}$$

$$\begin{aligned} AA^T &= (X^T X)^{-1} X^T (X^T X)^{-1} X^T \\ &= (X^T X)^{-1} X^T X (X^T X)^{-1} = (X^T X)^{-1} \end{aligned}$$

$$DA^T = D \left((X^T X)^{-1} X^T \right)^T = \underbrace{DX}_0 (X^T X)^{-1}$$

$$AD^T = (DA^T)^T = 0^T = 0 \quad \text{so}$$

$$GG^T = DD^T + (X^T X)^{-1}$$

$$GG^T - (X^T X)^{-1} = DD^T$$

now we have something for

$$GG^T - (X^T X)^{-1} \geq 0$$

we have shown that

$$\text{var}(\hat{\beta}) = (X^T X)^{-1} \sigma^2 \quad \text{for BLUE and}$$

$$\text{var}(\tilde{\beta}) = GG^T \sigma^2 \quad \text{for generic unbiased linear estimator.}$$

$\text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}) = (bb^T - (X^T X)^{-1}) \sigma^2$ since $\sigma^2 > 0$
to show $\hat{\beta}$ is Best, we want to show

$$bb^T \geq (X^T X)^{-1} \text{ or } bb^T - (X^T X)^{-1} \geq 0$$

we have just shown

$$bb^T - (X^T X)^{-1} = DD^T \geq 0$$

$$bb^T - (X^T X)^{-1} = DD^T = D^2 \geq 0$$

so D is positive semi-definite meaning
the minimum it can be is zero. So the Best
outcome for a general unbiased linear estimator
is if $DD^T = 0$ or $D = 0 \Rightarrow b = (X^T X)^{-1}$

$\therefore \hat{\beta}$ is the Best Unbiased Linear estimator of β

PROBLEM #2 } LINEAR REGRESSION w/ ORTHOGONAL DESIGN

Assume that the columns $x_0, x_1, \dots, x_p$ of X are orthogonal

$$X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{12} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix} \quad \langle \text{col}(X_i), \text{col}(X_j) \rangle = 0$$

Express $\hat{\beta}_j$ in terms of $x_0, x_1, \dots, x_p$ and y .

$$\Rightarrow y = X\beta + \varepsilon \quad \text{and} \quad \hat{y} = X\hat{\beta}$$

$$\text{For BLUE: } \hat{\beta} = (X^T X)^{-1} X^T y$$

$$X^T y = X^T X \hat{\beta}$$

What is $X^T X$?

$$X^T X = \begin{bmatrix} \text{---} & x_0 & \text{---} \\ \text{---} & x_1 & \text{---} \\ & \vdots & \\ \text{---} & x_p & \text{---} \end{bmatrix} \begin{bmatrix} | & | & & | \\ x_0 & x_1 & \dots & x_p \\ | & | & & | \end{bmatrix} \quad \begin{array}{l} p \text{ columns} \\ n \text{ rows} \end{array}$$

$$= \begin{bmatrix} x_0^T x_0 & x_0^T x_1 & \dots & x_0^T x_p \\ x_1^T x_0 & x_1^T x_1 & \dots & x_1^T x_p \\ \vdots & \vdots & \ddots & \vdots \\ x_p^T x_0 & x_p^T x_1 & \dots & x_p^T x_p \end{bmatrix}$$

if we assume orthogonality: $\langle x_i, x_j \rangle = x_i^T x_j = \delta_{ij}$
 $X^T X$ is a diagonal matrix.

$$X^T X = \begin{bmatrix} x_0^T x_0 & 0 & \dots & 0 \\ 0 & x_1^T x_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & x_p^T x_p \end{bmatrix}$$

so for $X^T X \hat{\beta}$, if columns of X are orthogonal, then

$$X^T X \hat{\beta} = \begin{bmatrix} x_0^T x_0 & 0 & \dots & 0 \\ 0 & x_1^T x_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & x_p^T x_p \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_n \end{bmatrix} = \begin{bmatrix} x_0^T x_0 \hat{\beta}_0 & 0 & \dots & 0 \\ 0 & x_1^T x_1 \hat{\beta}_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & x_p^T x_p \hat{\beta}_n \end{bmatrix}$$

what is $X^T y$? $X^T \in \mathbb{R}^{(p+1) \times n}$ $y \in \mathbb{R}^{n \times 1}$

$$X^T y = \begin{bmatrix} \text{--- } x_0 \text{ ---} \\ \text{--- } x_1 \text{ ---} \\ \vdots \\ \text{--- } x_p \text{ ---} \end{bmatrix} \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} x_0^T y \\ x_1^T y \\ \vdots \\ x_p^T y \end{bmatrix}$$

so for $X^T y = X^T X \hat{\beta}$ we have

$$\begin{bmatrix} x_0^T y \\ x_1^T y \\ \vdots \\ x_p^T y \end{bmatrix} = \begin{bmatrix} x_0^T x_0 \hat{\beta}_0 & 0 & \dots & 0 \\ 0 & x_1^T x_1 \hat{\beta}_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & x_p^T x_p \hat{\beta}_p \end{bmatrix}$$

$X \in \mathbb{R}^{n \times (p+1)}$ so $x_j = \text{col}(X) \in \mathbb{R}^{n \times 1}$ and $x_j^T \in \mathbb{R}^{1 \times n}$
and $y \in \mathbb{R}^{n \times 1}$ from $y = X\beta + \epsilon$. so $x_j^T y$ is actually
a scalar (inner product)

so we have

$$x_j^T y = x_j^T x_j \hat{\beta}_j \quad ; \quad \hat{\beta}_j = \frac{x_j^T y}{x_j^T x_j} = \frac{\langle x_j, y \rangle}{\langle x_j, x_j \rangle}$$

x_j is col vector $\in \mathbb{R}^{n \times 1}$
and y is col vector $\in \mathbb{R}^{n \times 1}$

given this formula, we can interpret $\langle x_j, y \rangle$ as the similarity of x_j and y while $\langle x_j, x_j \rangle$ is acting like a normalization of some sort.

PROBLEM 3 | THE MINIMUM NORM SOLUTION

When $X^T X$ is not invertible, the normal equations $X^T X \beta = X^T y$ do not have a unique solution.

Assume that $X \in \mathbb{R}_r^{n \times (p+1)}$, where r is the rank of X . Assume the SVD of X is $U \Sigma V^T$, where $U \in \mathbb{R}^{n \times r}$ satisfies $U^T U = \mathbb{I}_r$. Also $V \in \mathbb{R}^{(p+1) \times r}$ satisfies $V^T V = \mathbb{I}_r$ and $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$ is the diagonal matrix of positive singular values.

(a) show that $\beta_{\text{mns}} = V \Sigma^{-1} U^T y$ is a solution to the normal equations.

(b) show that for any other solution to the normal equations, $\|\beta\| \geq \|\beta_{\text{mns}}\|$. [Hint: one way (and not the only way) of doing this is to show that $\beta = \beta_{\text{mns}} + b$]

(c) Is $V \Sigma^{-1} U^T$ the pseudo-inverse of X ?

[Hint: you can prove or disprove this using the so-called penrose properties].

$\Rightarrow$ if $X^T X$ is not invertible, then $(X^T X)^{-1}$ does not exist so $\hat{\beta} = (X^T X)^{-1} X^T y$ is not a solution. need to solve $\hat{\beta} X^T X = X^T y$ some other way.

(a) let's see if $\beta_{\text{mns}} = V \Sigma^{-1} U^T y$ solves $X^T y = X^T X \beta$

if X has an SVD then

$$X = U \Sigma V^T; \quad X^T = V \Sigma U^T$$

$$X^T X = V \Sigma U^T U \Sigma V^T = V \Sigma \Sigma V^T = V \Sigma^2 V^T$$

so

$$X^T X \beta_{\text{mns}} = V \Sigma \Sigma V^T \cdot V \Sigma^{-1} U^T y \quad \text{using } V^T V = I$$

$$= V \Sigma \Sigma^{-1} U^T y$$

$$= V \Sigma U^T y$$

$$\text{using } X^T = V \Sigma U^T$$

$$= X^T y$$

$\therefore \beta_{\text{mns}}$ satisfies normal equations

(b) we want to show $\| \beta \| \geq \| \beta_{\text{mns}} \|$. we want to show that the norm of β is larger or equal to the norm of β_{mns} for any other solution to the normal eqns $\Rightarrow$ we do not know the form β will take when $(X^T X)^{-1}$ is not invertible, so per hint, we take $\beta = \beta_{\text{mns}} + b$. we will see what happens to this b .

$$\| \beta \| = \sqrt{\beta^T \beta} \quad \text{w } \beta = \beta_{\text{mns}} + b$$

$$\beta^T \beta = (\beta_{\text{mns}} + b)^T (\beta_{\text{mns}} + b)$$

$$= (\beta_{\text{mns}}^T + b^T) (\beta_{\text{mns}} + b)$$

$$= \beta_{mns}^T \beta_{mns} + \beta_{mns}^T b + b^T \beta_{mns} + b^T b$$

$$\|\beta\|^2 = \|\beta_{mns}\|^2 + \|b\|^2 + \beta_{mns}^T b + b^T \beta_{mns}$$

the cross terms ($\beta_{mns}^T b$ & $b^T \beta_{mns}$) could be negative. we need to show $\beta_{mns} \perp b$ so that their product is zero.

* after consulting chat gpt, I have made sense of this *

$$\text{we have } \|\beta\|^2 = \|\beta_{mns}\|^2 + \|b\|^2 + \beta_{mns}^T b + b^T \beta_{mns}$$

this almost shows $\|\beta\|^2 \geq \|\beta_{mns}\|^2$, but the cross terms may be negative, insinuating the opposite.

$\Rightarrow$ we used $\beta = \beta_{mns} + b \Rightarrow b = \beta - \beta_{mns}$. To

obtain more info on b , let's see how it acts on X .

$$Xb = X(\beta - \beta_{mns})$$

$$= X\beta - X\beta_{mns}$$

X has rank n . β_{mns} is built from SVD of X . Since $(X^T X)^{-1}$ is not invertible, X does not have full rank. Its diminished rank is β_{mns} has rank r , which is in the column space of X . The null space rank of X is $n-r$.

so $X\beta$ and $X\beta_{mns}$ are the same since though they have different ranks, their rank difference is in the null space

$$X\beta - X\beta_{mns} = X\beta - X\beta + 0 = 0 \quad \text{so}$$

$$Xb = X(\beta - \beta_{mns}) = X\beta - X\beta_{mns} = 0$$

* so if $Xb=0$ then $b \in \text{Null}\{X\}$ *

rank $n-r$

rank r

now, $b \in \text{Null}\{X\}$; $\beta_{mns} \in \text{Col}\{X\}$, so our equation

$$\|\beta\|^2 = \|\beta_{mns}\|^2 + \|b\|^2 + \beta_{mns}^T b + b^T \beta_{mns}$$

has cross terms equal to zero since $\beta_{mns} \in \text{Col}(X)$, $b \in \text{Null}(X)$ and $b \perp \beta_{mns}$ since column space & null spaces are orthogonal! Therefore

$$\|\beta\|^2 = \|\beta_{mns}\|^2 + \|b\|^2$$

$\|b\|^2 > 0$ since null space is non-empty hence

$$\boxed{\|\beta\|^2 \geq \|\beta_{mns}\|^2}$$

(c) We didn't cover the pseudo-inverse in EE503.

The following is my understanding from google & chatgpt.

$X\beta = y$ does not have a solution. pseudo inverse is represented as X^+ & is the least squares soln - it finds the value of X that minimize $\|X\beta - y\|$. If multiple solns exist, it picks the minimum vector.

$\Rightarrow$ to find it, you take the SVD of X

and take Σ^{-1} . $\Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r)$ so

$$\Sigma^{-1} = \text{diag}\left(\frac{1}{\sigma_1}, \frac{1}{\sigma_2}, \dots, \frac{1}{\sigma_r}\right)$$

$\Rightarrow$ in our problem, $\beta_{\text{mns}} = V \Sigma^{-1} U^T y$ and $\beta = X^+ y$ show that

$V \Sigma^{-1} U^T$ is the pseudo inverse of X . we show this using their definitions:

1. Reflexive property : $X X^+ X = X$
2. reverse reflexive : $X^+ X X^+ = X^+$
3. Symmetry of $X X^+$: $(X X^+)^T = X X^+$
4. Symmetry of $X^+ X$: $(X^+ X)^T = X^+ X$

to do this, we know X has an SVD of

$X = U \Sigma V^T$ and $X^\dagger = V \Sigma^{-1} U^T$ is its pseudoinverse.
Then:

$$\begin{aligned} 1. \quad X X^\dagger X &= U \Sigma V^T V \Sigma^{-1} U^T X & V^T V &= I \\ &= U \Sigma \Sigma^{-1} U^T X \\ &= U U^T X \\ &= U U^T U \Sigma V^T & U^T U &= I \\ &= U \Sigma V^T \\ &= X \quad \checkmark \end{aligned}$$

$$\begin{aligned} 2. \quad X^\dagger X X^\dagger &= V \Sigma^{-1} U^T U \Sigma V^T V \Sigma^{-1} U^T & U^T U &= V^T V = I \\ &= V \Sigma^{-1} \Sigma \Sigma^{-1} U^T & \Sigma^{-1} \Sigma &= \Sigma \Sigma^{-1} = I \\ &= V \Sigma^{-1} U^T \\ &= X^\dagger \quad \checkmark \end{aligned}$$

$$\begin{aligned} 3. \quad (X X^\dagger)^T &= (U \Sigma V^T V \Sigma^{-1} U^T)^T & V^T V &= I \\ &= (U \Sigma \Sigma^{-1} U^T)^T & \Sigma \Sigma^{-1} &= I \\ &= (U U^T)^T \\ &= U U^T \end{aligned}$$

since $(U U^T)^T = U U^T$, $U U^T$ is symmetric. if
 $X X^\dagger = U U^T$ then $X X^\dagger$ is symmetric $\checkmark$

$$\begin{aligned}
 4. \quad (X^T X)^T &= (V \tilde{Z}^T U^T U \tilde{Z} V^T)^T & U^T U &= I \\
 &= (V \tilde{Z}^T \tilde{Z} V^T)^T & \tilde{Z}^T \tilde{Z} &= I \\
 &= (V V^T)^T \\
 &= V V^T
 \end{aligned}$$

since $(V V^T)^T = V V^T$ then $V V^T$ is symmetric. If $X^T X = V V^T$, by transitive property $X^T X$ is also symmetric. ✓